

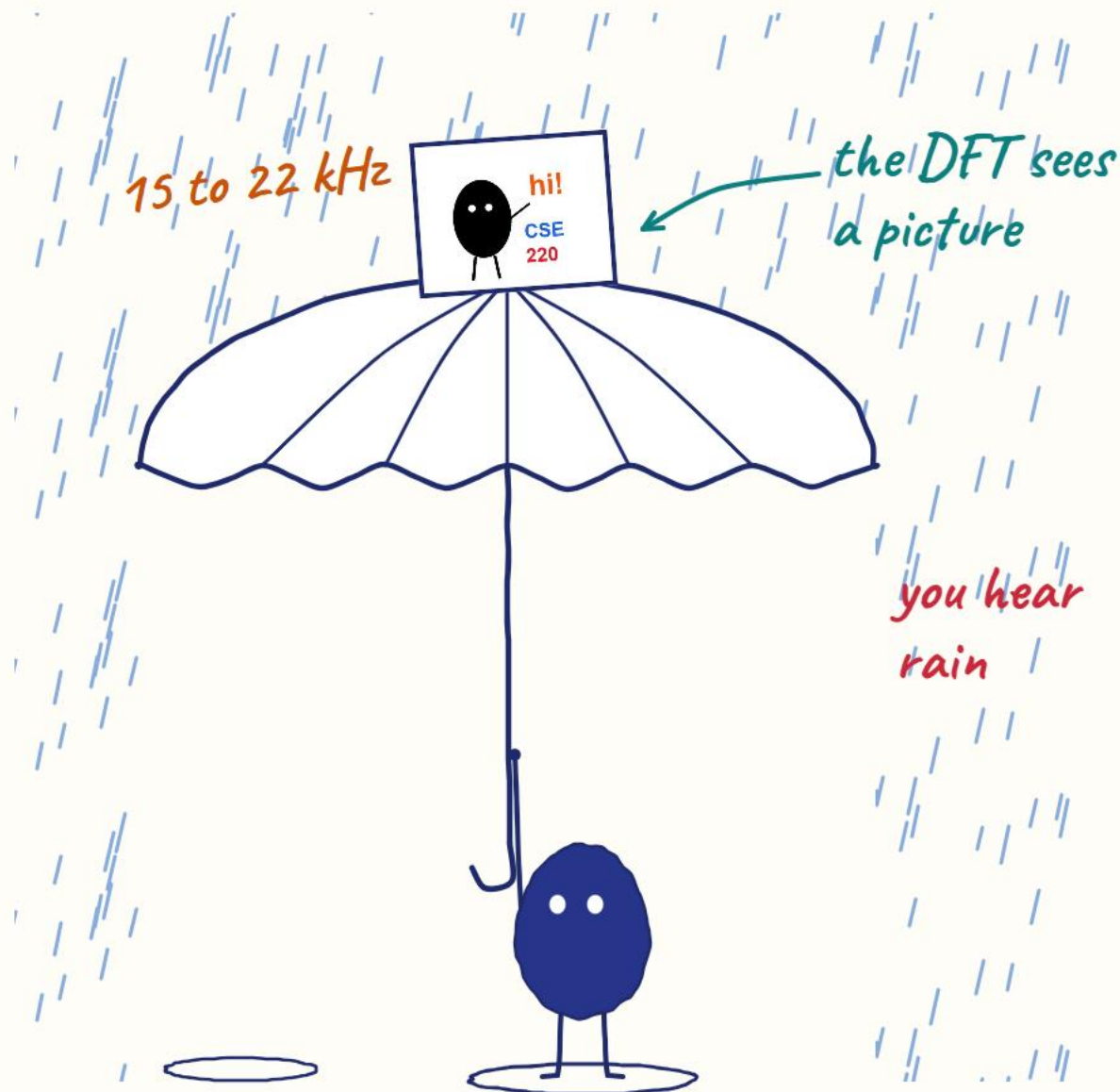
HiddenHz

A picture hidden in the rain

Rayyan Khalil · 2305098

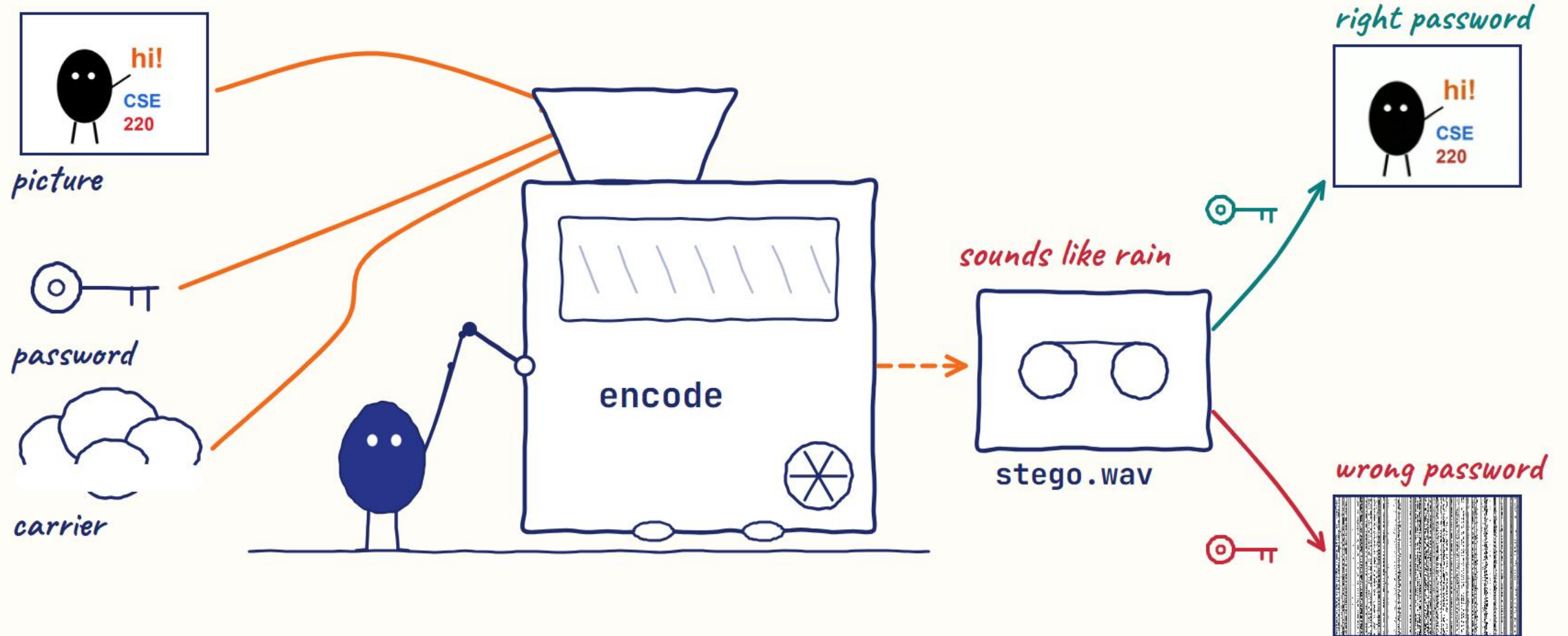
Md. Iztiyah Islam · 2305095

CSE 220 · Signals and Linear Systems



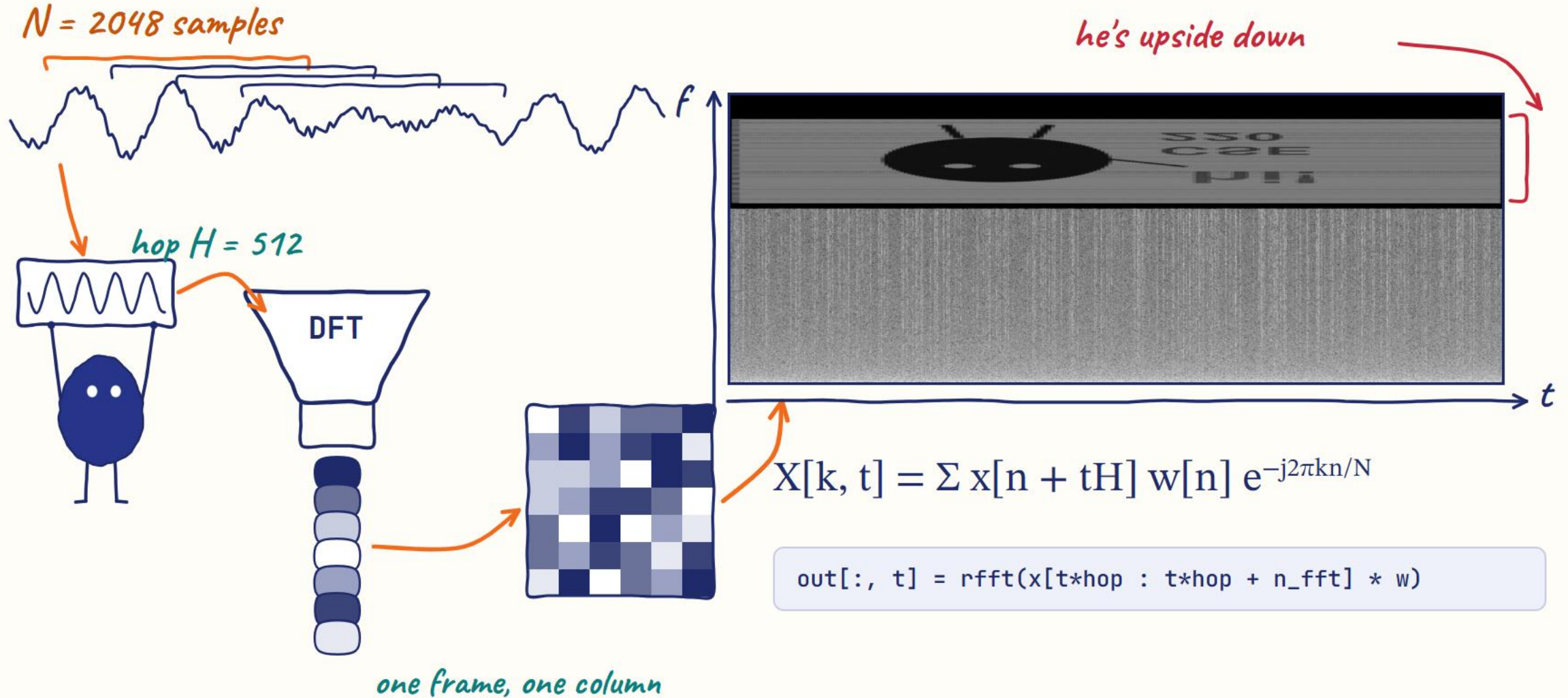
THE IDEA

How it works



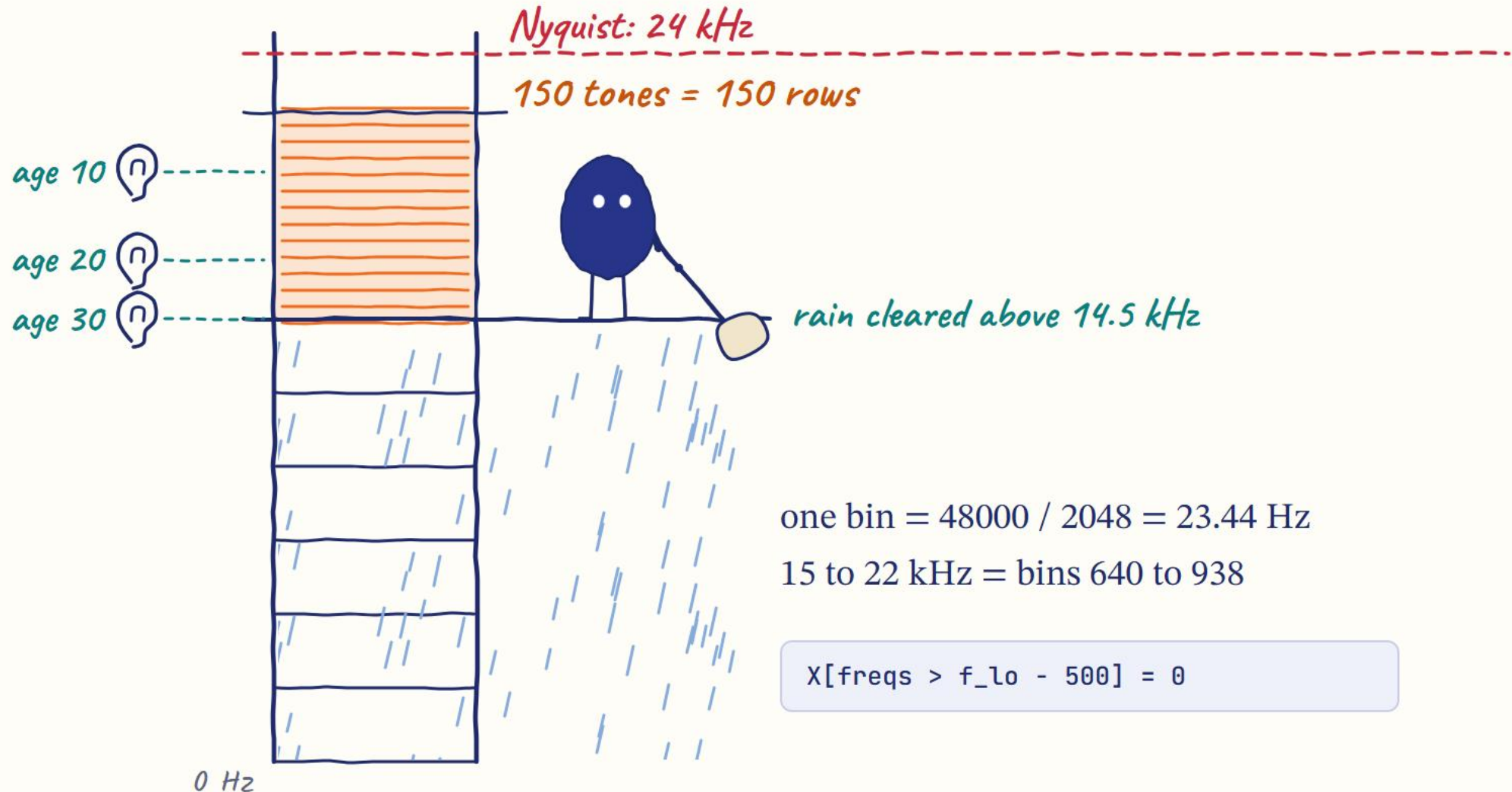
THEORY

A spectrogram is an image



THEORY

Where the picture lives: 15 to 22 kHz



THEORY

Hann window: three bins per tone

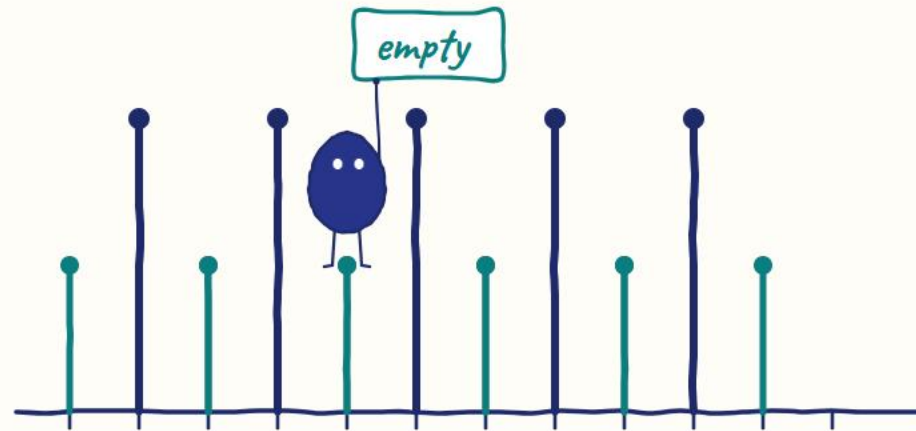
$$w[n] = 0.5 - 0.5 \cos(2\pi n / N)$$
$$= 0.5 - 0.25 e^{+j2\pi n/N} - 0.25 e^{-j2\pi n/N}$$

```
w = 0.5 - 0.5*np.cos(2*np.pi*np.arange(n)/n)
```



a chopped tone leaks everywhere

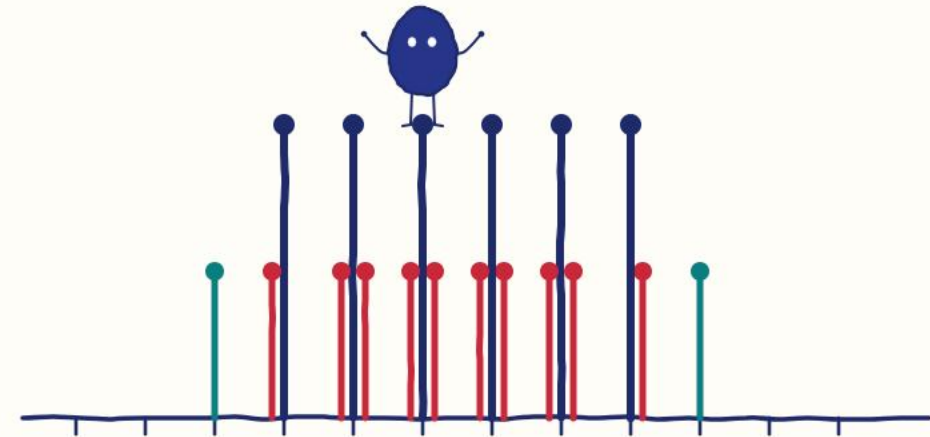
spacing 2 (ours)



quality 0.98

HiddenHz

spacing 1

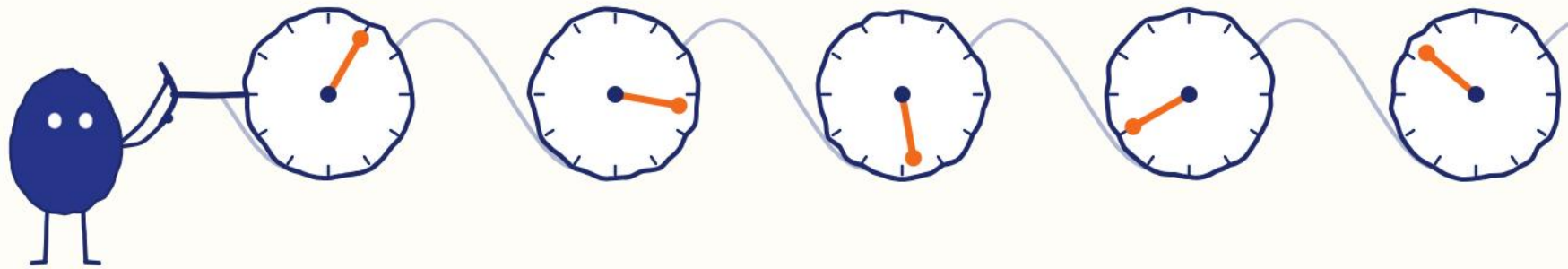


quality 0.59

THEORY

Phase that advances like a real tone

$$\varphi[k, t] = \varphi_0[k] + 2\pi k H t / N$$



*same step
every hop*



*random:
frames disagree*

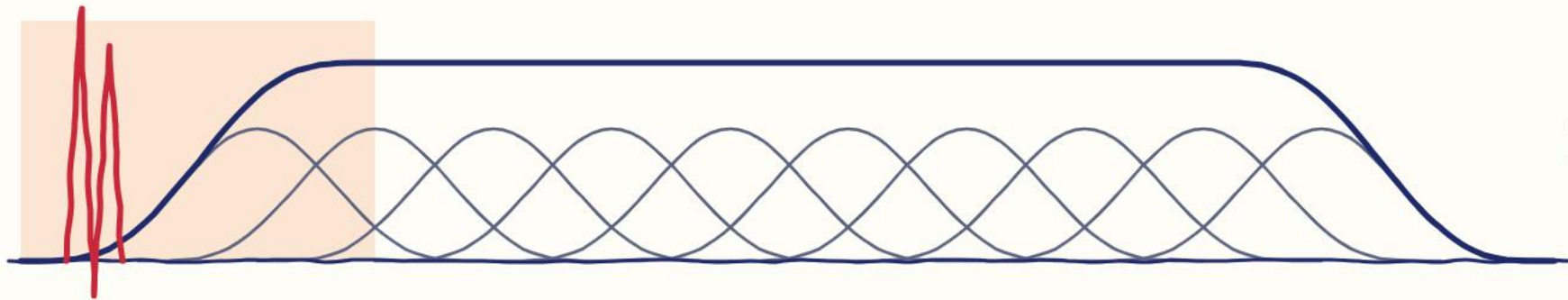
```
phase = phi0 + 2*np.pi * bins[:, None] * hop * t / n_fft
```

THEORY

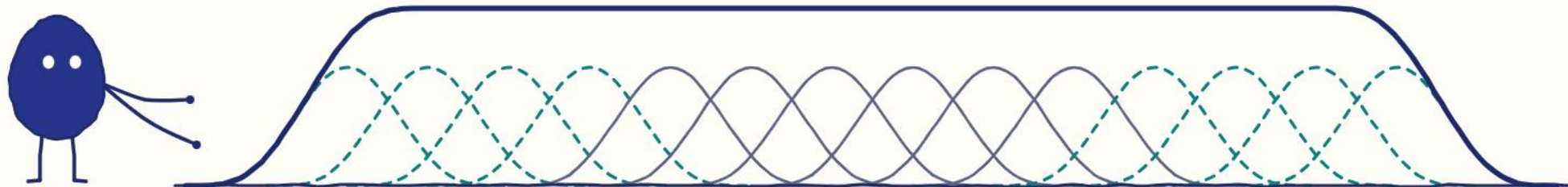
Back to sound: the inverse STFT

$$y[n] = \sum_t w[n - tH] \cdot \text{IDFT}_t[n - tH] / \sum_t w^2[n - tH]$$

click:
14.2 % of
the energy



4 blank
frames

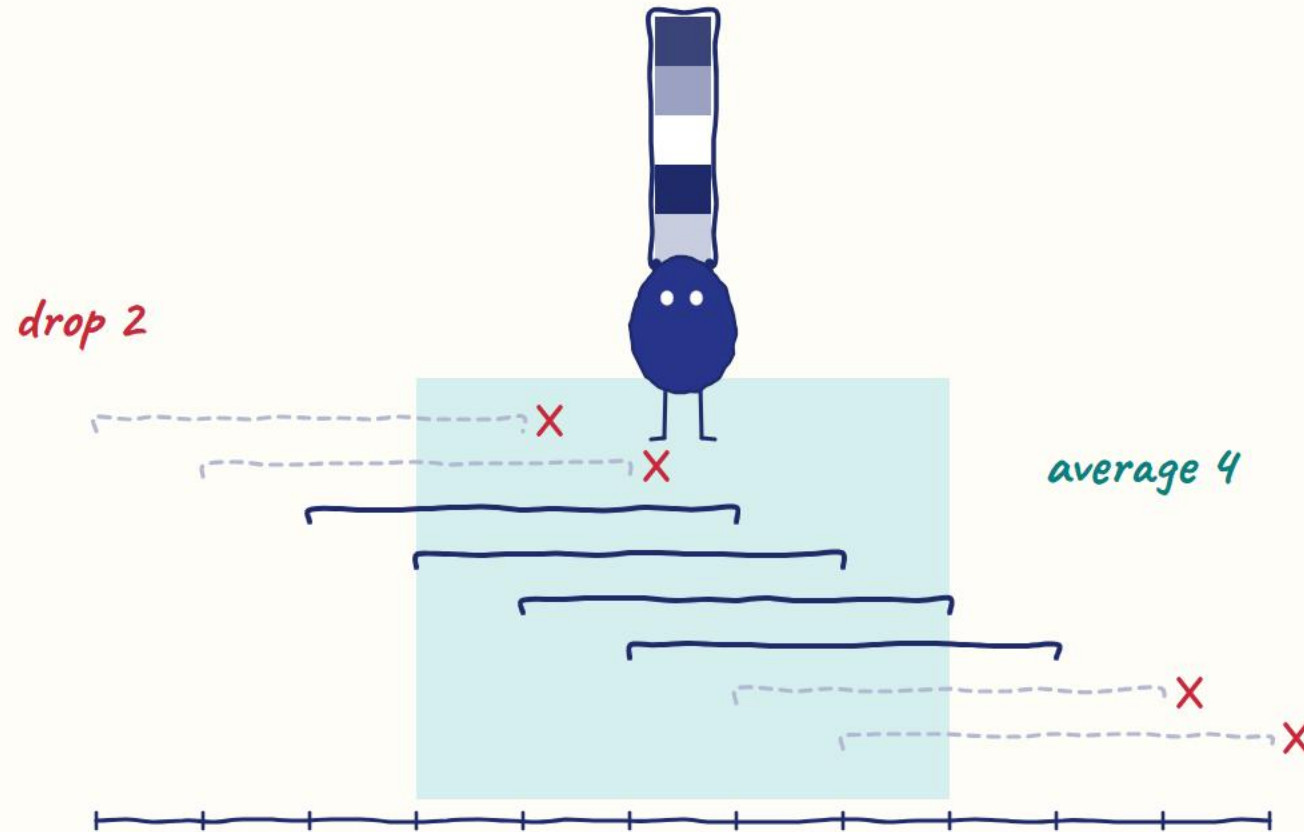


```
pad = np.zeros((rows, 4))
```

THEORY

Each column lasts eight frames

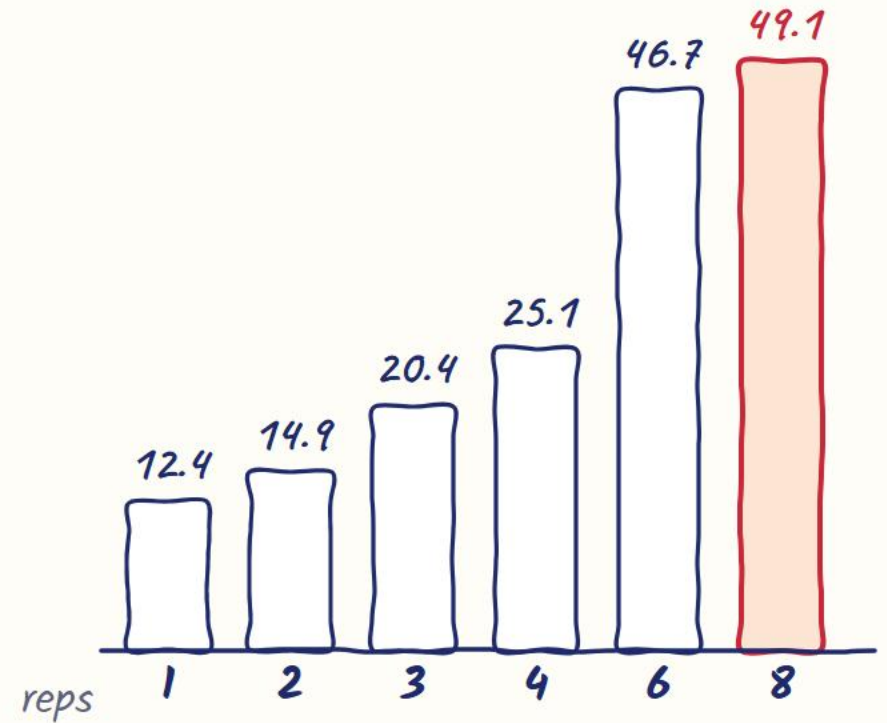
$$\text{reps} - 3 \geq 4 \rightarrow \text{reps} \geq 7$$



```
mag[:, :, 2:6].mean(axis=2)
```

HiddenHz

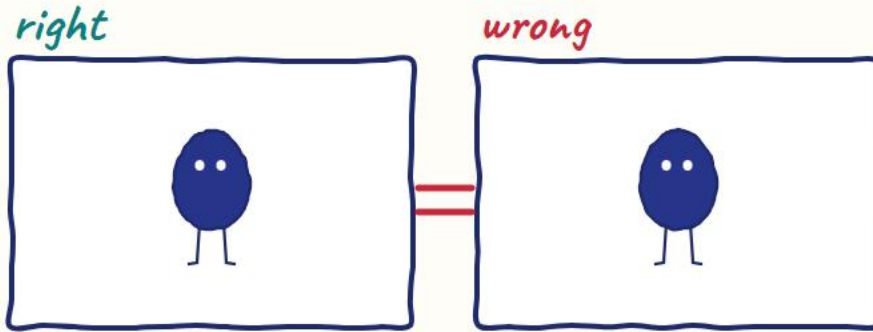
PSNR (dB)



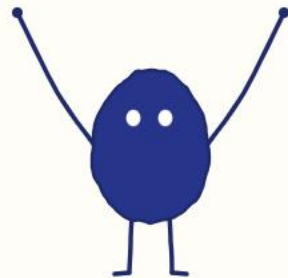
SECURITY

What the password does

random phase?

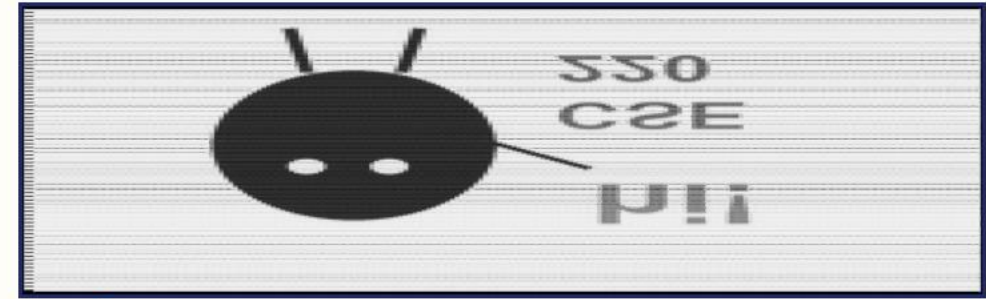


$$0.914 = 0.914$$



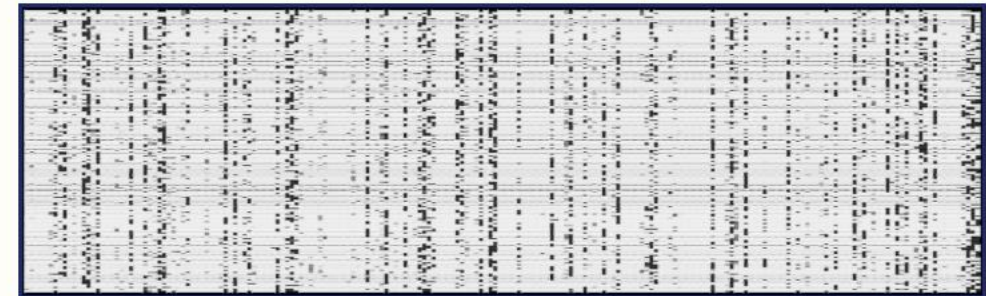
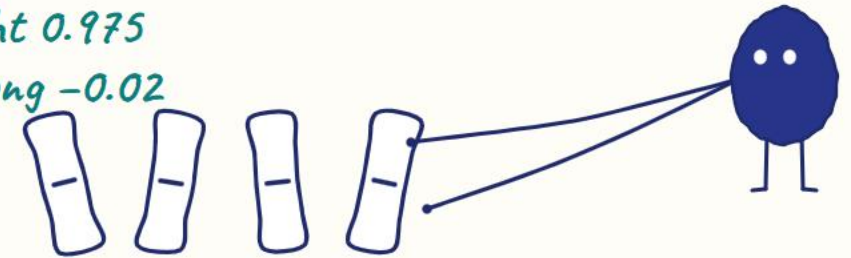
a spectrogram ignores phase

shuffle rows and columns



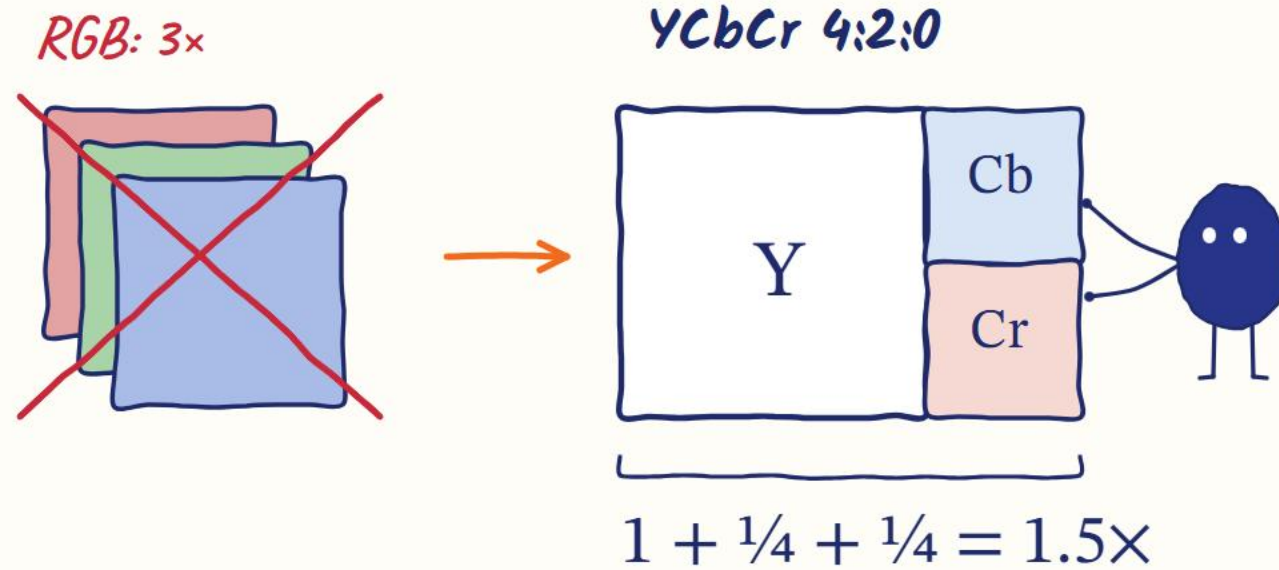
right 0.975

wrong -0.02



```
rows, cols, phi0 = key_schedule(pw, R, C)
```

Capacity and colour



length $\approx 2 R^2 \cdot \text{spacing} / \text{bandwidth}$

150 × 150: 13.1 s grey, 19.5 s colour

files: bits + Hamming(7,4) + CRC-32



≈ 125 bytes per second

The app: encoding

Hide an image in sound

HIDE

Image File
PDF, ZIP...

IMAGE

 secret.png
13 KB REPLACE

CARRIER OPTIONAL

 rain.wav
3.7 MB REPLACE

Any common format (MP3, M4A, WAV...), converted on the server. 44.1 or 48 kHz keeps the most detail. Without one, the band alone.

☒ RECORD CARRIER

Or capture from your microphone. Saved as audio and converted on the server.

PASSWORD

..... SHOW

At least 4 characters. The same password decodes it.

COLOUR PRESET

☒ ON Standard 150 rows Detail 299 rows

ROWS SQUARE IMAGE At 48 kHz. A 44.1 kHz carrier gives 116 rows. Colour costs 1.5x grayscale.

150 195 s

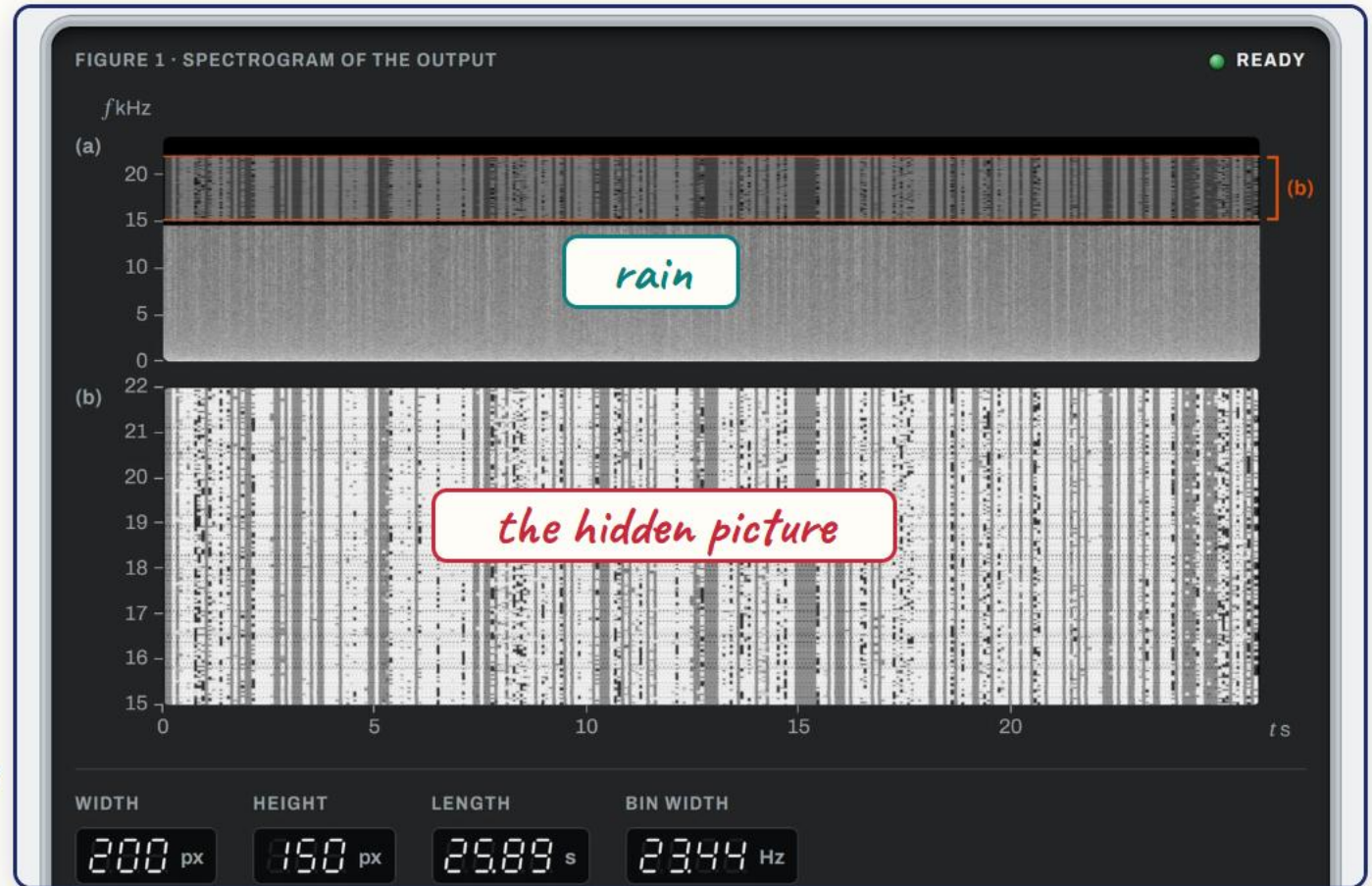
☒ Hide image in audio

← image or file

← any audio

← or record

← colour, detail



Recording and hiding files

record a carrier

● STOP RECORDING 0:03

Or capture from your microphone. Saved as audio and converted on the server.

hide a small PDF

Hide a file in sound

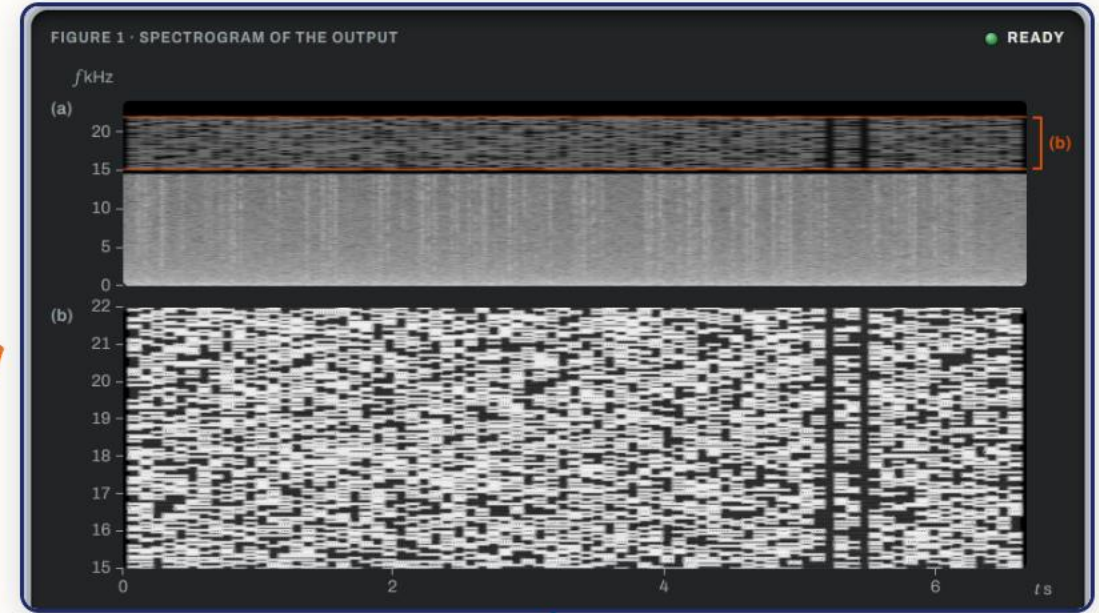
HIDE

ImageFile
PDF, ZIP...

FILE

lab-notes.pdf
766 BREPLACE

Any file up to about 70 KB (a short PDF, a text file, a small ZIP), recovered byte-for-byte. MP3 or re-recording the audio destroys it.



● File recovered

lab-notes.pdf
766 bytes · CRC verified

Download file

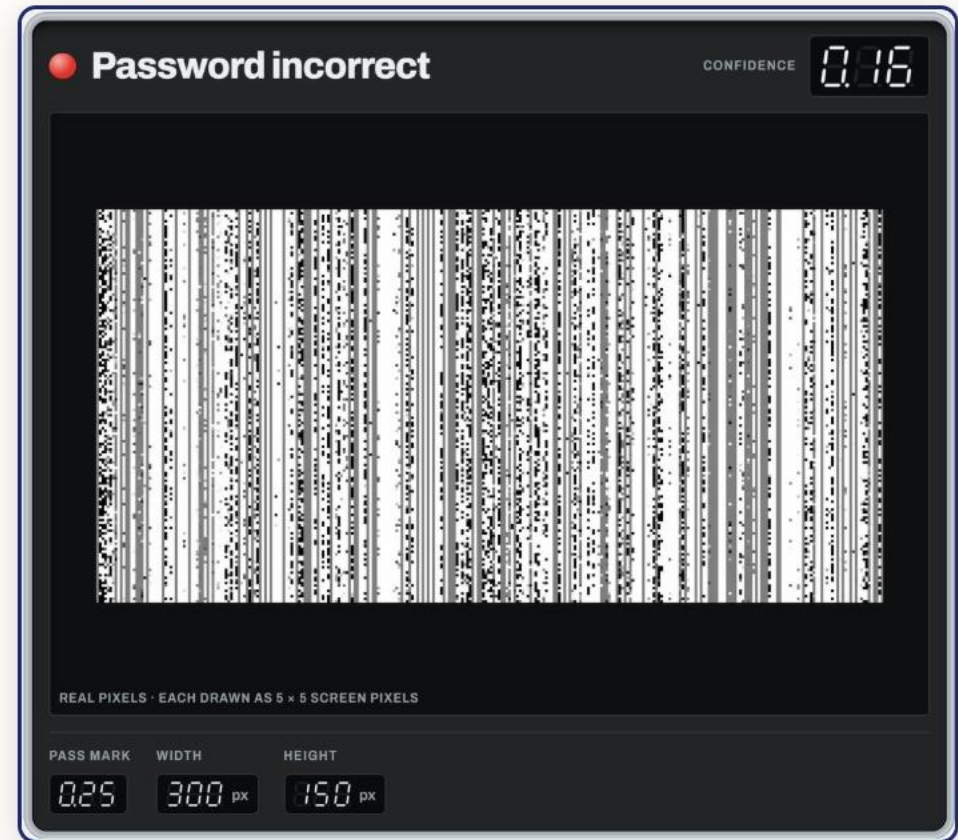
byte for byte, CRC checked

The app: decoding

right password

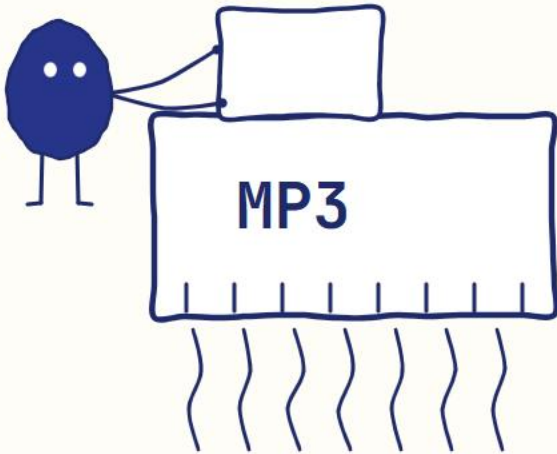


wrong password

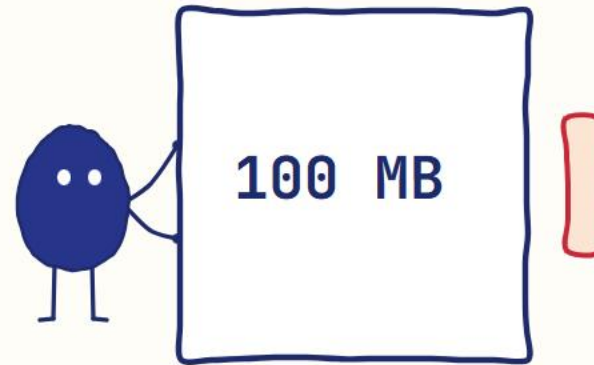


LIMITS

Limitations



*MP3 and AAC remove
the band. Keep WAV.*



≈ 125 bytes / s

70 KB: 10 minutes

5 MB PDF: 12 hours

100 MB video: 10 days



*anyone can see
something is there*

Thank you

Questions?

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CSE 220 · Signals and Linear Systems

